

Perturbation-Resilient Autonomous Navigation with Distributionally Robust Reinforcement Learning

Zhaofan Zhang¹, Minghao Yang¹, Sihong Xie^{1*} and Hui Xiong^{1,2*}

¹Thrust of Artificial Intelligence, The Hong Kong University of Science and Technology (Guangzhou), China

²Department of Computer Science and Engineering, The Hong Kong University of Science and Technology Hong Kong SAR, China

{zzhang398, myang272}@connect.hkust-gz.edu.cn, sihongxie@hkust-gz.edu.cn, xionghui@ust.hk

Abstract

The robustness of autonomous vehicles such as drones and Unmanned Surface Vehicles (USV) is crucial when facing unknown and complex marine environments, especially when heteroscedastic observational noise poses significant challenges to sensor-based navigation tasks. Recently, Distributional Reinforcement Learning (DistRL) has shown promising results in some challenging autonomous navigation tasks without prior environmental information. However, these methods overlook situations where noise patterns vary across different environmental conditions, hindering safe navigation and disrupting the learning of value functions. To address the problem, we propose DRIQN to integrate Distributionally Robust Optimization (DRO) with implicit quantile networks to optimize worst-case performance under natural environmental conditions. Leveraging explicit subgroup modeling in the replay buffer, DRIQN incorporates heterogeneous noise sources and target robustness-critical scenarios. Experimental results based on the risk-sensitive environment demonstrate that DRIQN significantly outperforms state-of-the-art methods, achieving +13.51% success rate, -12.28% collision rate and +35.46% for time saving, +27.99% for energy saving, compared with the runner-up.

1 Introduction

Autonomous navigation is a foundational capability for unmanned systems, enabling both USVs and drones to transition from teleoperation to autonomous, goal-directed operation under uncertain and rapidly changing conditions. Reliable navigation empowers vehicles to undertake long-duration missions ranging from port oil spill cleanup [Elmakis and Degani, 2023] and environmental monitoring [Maharjan *et al.*, 2022] to border security [Fernandes *et al.*, 2025] and search-and-rescue [Wang *et al.*, 2023b] while simultaneously reducing operational risk, human exposure, and overall cost. In these navigation tasks, the environment is characterized by complexity and partial observability. The intricate

*Corresponding authors.

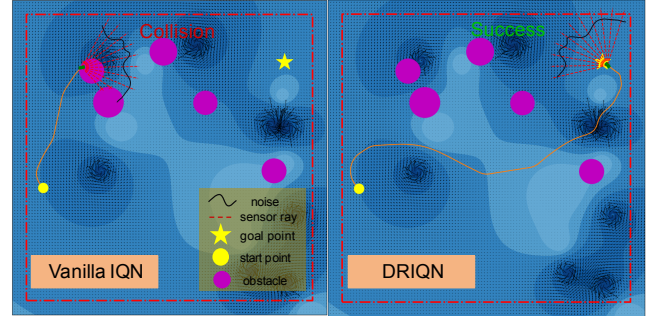


Figure 1: **IQN vs. Robust DRIQN**: USV navigation trajectories visualization with observational noise. Vanilla IQN may be biased by noisy observations, resulting in risky policies.

variations in water currents and airflow are difficult to monitor in real-time, and obstacles may also be encountered [Lolla *et al.*, 2015]. Moreover, the perception devices used by the robot may face various unobservable interferences within the environment, resulting in noisy observations [Vieira *et al.*, 2020]. Specifically, heterogeneous extreme natural conditions (e.g., weather, sea state, geographic locality) induce trajectory-specific, non-stationary observational noise. As illustrated in Figure 1, the observational perturbation exacerbates the inherent uncertainty of complex marine environments and impedes accurate risk-sensitive policy learning, creating substantial navigation hazards for unmanned vehicles, whereas our proposed DRIQN remains robust to such corruption and progresses reliably toward the target.

Prior works on autonomous navigation can be mainly grouped into: (i) *model-based* methods [Wang *et al.*, 2023a; Asfihani *et al.*, 2025]; (ii) models with *prior-dependent* knowledge [Han *et al.*, 2016; Hong *et al.*, 2024]; (iii) *direct observation-driven decision-making* approaches (e.g., APF [Sun *et al.*, 2019], BA [McGuire *et al.*, 2018] and reinforcement learning [Lin *et al.*, 2023; Wang *et al.*, 2024]). Direct observation-driven approaches make decisions based on sensor observation without prior environmental knowledge and extra modeling design. Despite promising results, model-based approaches (e.g., nonlinear MPC) incorporate analytical vehicle dynamics into optimal control, requiring high-fidelity models and suffering degradation under uncer-

ainties. Prior-dependent methods further face limitations in prior availability and freshness, incurring substantial precomputation overhead. In contrast, observation-driven decision-making methods offer a generalizable, prior-independent solution for navigation in unknown environments, albeit with heightened sensitivity to observational noise. Recently, as one kind of prominent observation-driven decision-making method, Deep Reinforcement Learning (DRL) has emerged for robotic navigation in complex environments [Hossain *et al.*, 2024]. Building on this advantage, some DistRL methods [Lin *et al.*, 2023; Liu *et al.*, 2023] enable risk-sensitive navigation through quantile-based return distribution modeling. By employing task-specific reward designs and distortion functions with Conditional Value-at-Risk (CVaR), DistRL reshapes return distributions to focus on critical lower-tail outcomes while constraining distributional support, thereby enhancing robust policy learning. Nevertheless, without environmental priors or modeling knowledge for constraint factors, these approaches cannot be readily optimized via constructing robust mechanisms, thereby lacking explicit safety guarantees under heterogeneous noise-corrupted observations. This presents a pressing challenge: how can we ensure the safety of unmanned surface navigation in multi-noise-corrupted marine environment just using a data-driven optimization approach?

To address the challenge, we propose Distributionally Robust Implicit Quantile Networks (DRIQN), a novel framework ensuring risk-sensitive policies via quantile regression under observational noise. Leveraging distributionally robust optimization, DRIQN reformulates the DRO problem as a semi-definite program (Eq. 13) to holistically address diverse noise influences. Crucially, DRIQN implements “gradient substitution” replacing the network’s output-layer gradients with DRO-based gradients to enhance robustness without auxiliary modules. Considering the scalability challenge of DRO caused by the large-scale datapoint-level optimization process, we transform standard datapoint-based DRO by leveraging transition data’s inherent structure: under specific environmental conditions, noisy transitions naturally form homogeneous subgroups following corresponding noise patterns. This structural property enables reformulating the original datapoint-level DRO problem as the supremum over subgroup-level loss functions. Each subgroup represents unlabeled transition data clusters with distinct noise characteristics. Experimental results demonstrate that DRIQN significantly outperforms baselines: achieving +13.51% success rate, -12.28% collision rate and +35.46% for time saving, +27.99% for energy saving, compared with the runner-up. To our knowledge, DRIQN is the first method tackling multiple observational noise patterns from natural conditions, enabling robust risk-sensitive policies for autonomous navigation.

The main contributions of this paper are as follows:

- We propose a novel, streamlined distributionally robust optimization framework that enhances RL agent performance against perturbed observations through substitution of DRO-based gradients.
- We present a risk-sensitive and robust navigation approach for developing an online path planner by unifying

DistRL and DRO, maintaining robustness and safety in the presence of complex environmental noise patterns.

- Simulated experiments show that the resulting planner remains robust to observation perturbations, enabling safe goal reaching under strong disturbances, unknown flows, and obstacles, and that this robustness generalizes across tasks and vehicle types.

2 Related Work

Among autonomous navigation paradigms (model-based, prior-dependent, and direct observation-driven), observation-driven methods are particularly suitable for planning in unknown environments due to their generality and minimal prior requirements. Classic representatives include Artificial Potential Field (APF) methods [Khatib, 1985; Sun *et al.*, 2019], which combine goal attraction and obstacle repulsion but suffer from local minima and goal-proximity failures, and Bug algorithms (e.g., Bug1/Bug2) [Lumelsky and Stepanov, 1986; McGuire *et al.*, 2018], which follow obstacle boundaries for reactive planning yet often yield highly suboptimal paths due to exhaustive perimeter traversal. More recently, deep RL has been widely adopted for navigation. For safety-critical settings, safe RL methods [Lütjens *et al.*, 2018; Dawood *et al.*, 2024; Tomilin *et al.*, 2025] enforce constrained optimization, but typically require handcrafted constraints and can still incur catastrophic failures under transient violations. However, adversarial robustness to synthetic perturbations and constraint-based Safe RL are not well aligned with our setting, which focuses on naturally occurring observation noise without manually imposed safety constraints. Without the aforementioned limitations, DistRL demonstrated superior performance in risk-sensitive tasks such as quadruped locomotion [Shi *et al.*, 2024a], pursuit-evasion [Zhang *et al.*, 2025], and autonomous navigation [Lin *et al.*, 2023; Liu *et al.*, 2023]. Nevertheless, severe observational noise induces a distributional shift between the learned and deployed returns. This mismatch compromises the reliability of standard distribution estimates.

For this non-trivial challenge, DRO can provide a principled framework for robust decision-making under uncertainty [Nietert *et al.*, 2023]. Specifically, DRO solves a min-max problem that minimizes the maximum expected loss over an uncertainty set of plausible data distributions centered on the empirical distribution [Staib and Jegelka, 2019]. Unlike expected-value optimization under domain randomization [Tobin *et al.*, 2017], which does not directly control worst-case degradation, DRO emphasizes adverse extremes, aligning more closely with navigation safety. Prior studies have leveraged DRO to improve robustness, but many introduce auxiliary components (e.g., generative models [Shi *et al.*, 2024b; Ramesh *et al.*, 2024]), model-based rule [Shi and Chi, 2024; He *et al.*, 2025], offline data [Leung *et al.*, 2025]). These requirements limit applicability to real-world problem, where agents must learn online in unknown environments without prior models or curated datasets.

3 Methodology

3.1 Problem Overview

An autonomous vehicle does not have full knowledge of the environment, but instead receives noisy partial observations of robot’s surroundings constrained by the limited sensing range of its sensors. We formulate the automated navigation problem as a *perturbed* Partially Observable Markov Decision Process $\mathcal{M}$, denoted as $(\mathcal{S}, \mathcal{A}, \mathcal{O}, \mathcal{P}, R, \gamma, \nu)$, where $\mathcal{S}$, $\mathcal{A}$ and $\mathcal{O}$ represent the state, action and observation space, respectively. At each time step t , the agent receives a corrupted observation $\nu(o_t)$ subject to natural disturbances, where $o_t \in \mathcal{O}$ denotes the observation of current state s_t . Based on this observation, the agent selects an action $a_t \in \mathcal{A}$ according to policy $\tilde{\pi}(a_t | \nu(o_t))$. This leads the agent to transition to the next state $s_{t+1} \sim \mathcal{P}(\cdot | s_t, a_t)$, and the agent receives a reward $r_t \sim R(s_t, a_t)$, as well as a perturbed observation $\nu(o_{t+1})$ of s_{t+1} . For observation-disturbed policy $\tilde{\pi}$, the discounted sum of future rewards is a random variable $Z^{\tilde{\pi}}(s_t, a_t)$, defined as $Z^{\tilde{\pi}}(s_t, a_t) = \sum_{k=0}^{\infty} \gamma^k R(s_{t+k}, a_{t+k})$, where $\gamma \in (0, 1)$ denotes the future reward discount factor. The objective of standard RL under noisy observations is to maximize the expected discounted return, formalized by the value function: $Q^{\tilde{\pi}}(s_t, a_t) = \mathbb{E}[Z^{\tilde{\pi}}(s_t, a_t)]$. Accordingly, expectations are risk-neutral, representing returns solely through $Q^{\tilde{\pi}}(s, a)$. DistRL addresses the limited expressiveness of this expectation by estimating the distribution of $Z^{\tilde{\pi}}(s, a)$. Environmental noise is incorporated into the state transition kernel to form the state representation.

3.2 Distributional Reinforcement Learning

Distributional Reinforcement Learning explicitly models the return distribution Z governed by the following *distributional* Bellman equation and optimality equation:

$$Z^{\pi}(s, a) \stackrel{D}{=} R(s, a) + \gamma Z^{\pi}(s', a'), \quad (1)$$

$$\mathcal{T}Z^{\pi}(s, a) \stackrel{D}{=} R(s, a) + \gamma Z^{\pi}\left(s', \arg\max_{a'} \mathbb{E}[Z^{\pi}(s', a')]\right), \quad (2)$$

where $\mathcal{T}$ represents the *distributional Bellman optimality operator*. Here $\stackrel{D}{=}$ indicates distribution equality.

We utilize DistRL with IQN as a risk-sensitive path planner that has achieved promising performance in many tasks of robotics [Liu *et al.*, 2023; Lin *et al.*, 2023; Shi *et al.*, 2024a]. IQN utilizes the quantiles to provide a flexible representation of return distribution Z , denoted as $Z_{\tau} := F_Z^{-1}(\tau)$, where $\tau \sim U([0, 1])$ and F^{-1} is inverse cumulative distribution function. To effectively capture risk-sensitive behaviors, IQN incorporates a distortion function β to adjust the quantile distribution, emphasizing certain regions of the return distribution. Then the distorted expectation of Z under $\beta : [0, 1] \rightarrow [0, 1]$ is defined by:

$$Q_{\beta}(s, a) = \mathbb{E}_{\tau \sim U([0, 1])}[Z_{\beta(\tau)}(s, a)]. \quad (3)$$

The standard deep Q -Learning framework trains a neural network-based state-action value function by iteratively minimizing a loss function, which is defined through the temporal difference (TD) error. The TD error is defined as:

$$\delta = r + \gamma \max_{a'} Q(s', a') - Q(s, a). \quad (4)$$

The IQN loss function integrates the sampled temporal difference error Eq. (6) with the *Huber* quantile regression [Huber, 1992] Eq. (7) to optimize the quantile value network:

$$\mathcal{L}_{IQN}(s, a, r, s') = \frac{1}{N'} \sum_{i=1}^N \sum_{j=1}^{N'} \rho_{\tau_i}^{\kappa}(\delta^{\tau_i, \tau'_j}) \quad (5)$$

where N and N' denote the counts of i.i.d. samples $\{\tau_i\}_{i=1}^N$ and $\{\tau'_j\}_{j=1}^{N'}$ from $\mathcal{U}([0, 1])$. The temporal difference error between quantile targets is defined as:

$$\delta^{\tau_i, \tau'_j} = r + \gamma Z_{\tau'_j}(s', a') - Z_{\tau_i}(s, a) \quad (6)$$

To mitigate the impact of outliers, we employ the Huber quantile loss:

$$\begin{aligned} \rho_{\tau}^{\kappa}(u) &= |\tau - \mathbb{I}_{\{u < 0\}}| \left(\frac{\mathcal{L}_{\kappa}(u)}{\kappa} \right) \\ \mathcal{L}_{\kappa}(u) &= \begin{cases} \frac{1}{2}u^2, & |u| \leq \kappa \\ \kappa(|u| - \kappa/2), & \text{otherwise} \end{cases} \end{aligned} \quad (7)$$

Minimizing the expectation of this loss yields a convex objective for the quantile-based method. Consequently, the greedy policy is approximated by averaging over K quantile samples $\tilde{\tau}_k \sim \mathcal{U}([0, 1])$:

$$\tilde{\pi}_{\beta}(s) = \arg\max_a \frac{1}{K} \sum_{k=1}^K Z_{\beta(\tilde{\tau}_k)}(s, a) \quad (8)$$

with actions selected through $\tilde{\pi}_{\beta}(s)$.

3.3 Navigation under Observational Perturbation

To evaluate navigation robustness, we utilize two distinct simulation environments: a drone navigating in cluttered indoor spaces and a USV navigating dynamic ocean scenarios. In both settings, the agent is tasked with reaching a target destination under natural observational noise and environmental disturbances. The RL setting is defined for the marine environment and is applied analogously to the drone environment with only minor adjustments following prior work [Liu *et al.*, 2022]: **Reward.** To encourage goal-directed motion and collision avoidance, we define $r_t = r_{\text{step}} + \alpha(d_{t-1} - d_t) + r_{\text{collision}}\mathbb{I}_{\text{collision}} + r_{\text{goal}}\mathbb{I}_{\text{goal}}$, where d_t is the distance to the goal at time t , and $\mathbb{I}_{\text{collision}}$ (resp. $\mathbb{I}_{\text{goal}}$) equals 1 if a collision occurs (resp. the goal is reached) at t , and 0 otherwise. We set $r_{\text{step}} = -1.0$, $r_{\text{collision}} = -50.0$, $r_{\text{goal}} = 100.0$, and $\alpha = 1.0$ following [Liu *et al.*, 2022]. **Observation.** Following real-world sensing, the observation at time t is $O_t = (O_{\text{velocity}}, O_{\text{goal}}, O_{\text{LiDAR}})$, where O_{velocity} is the seafloor-relative velocity, O_{goal} is the goal position in the robot frame, and O_{LiDAR} denotes the LiDAR return vector that captures obstacle reflections within 10m over a $2\pi/3$ rad field of view. **Action.** The action at time t is $a_t = (a(t), w(t))$, where $a(t)$ controls the change rate of speed magnitude and $w(t)$ controls the change rate of heading direction. The discrete action set is $a \in \{-0.4, 0.0, 0.4\} m/s^2$ and $w \in \{-0.52, 0.0, 0.52\} rad/s$, with forward speed constrained to $[0, v_{\text{max}}]$. **Observational Perturbation.** To model natural noise induced by meteorological, illumination, and environmental factors (rather than adversarial attacks), we use non-synthetic noise sources specified in Section 4.1.

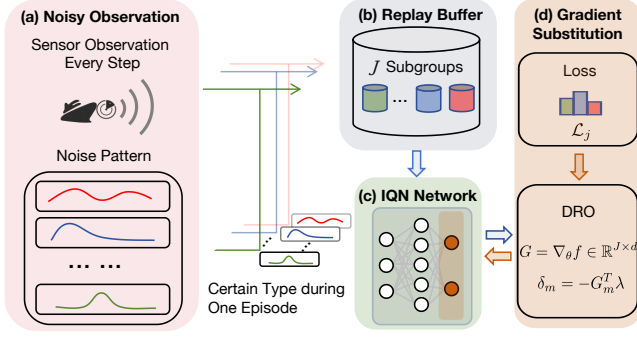


Figure 2: **Overall framework of our method.** In (a), the USV receives noisy observations specific to its current natural condition, assuming a consistent noise pattern within one episode due to proximity of environmental condition. Therefore, the replay buffer in (b) accumulates subgroups with distinct noise patterns. This variation has a significant impact on the training of the implicit quantile networks in (c). To enhance robustness, DRIQN employs “Gradient Substitution” in (d), replacing the gradients of the output layer with those computed via DRO.

3.4 DRO for Online DistRL

IQN has demonstrated superior performance as a risk-sensitive policy for path planning in an ideal environment. However, the planner relies on online distributional reinforcement learning training without prior information, meaning that errors in actions under noisy observations will cause reward signal mismatching and policy deviation. This issue is exacerbated by the presence of multiple noise subgroups within the replay buffer. Despite its impact, this problem remains largely unexplored in existing research.

To address heterogeneous noise in robot sensor observations, we propose DRIQN, as shown in Figure 2. DRIQN casts robustness under noisy transition data as an optimization problem and incorporates DRO during training.

Based on prior work on DRO theory [Qi *et al.*, 2020; Shen *et al.*, 2022], we further integrate the DRO objective into IQN training by formulating the overall optimization target as:

$$\min_{\theta \in \mathbb{R}^d} F_p(\theta) = \min_{\theta \in \mathbb{R}^d} \sup_{\mathbf{P} \in \mathbb{R}^n} \sum_{i=1}^n p_i \mathcal{L}_{IQN}(T_i) - d(\mathbf{P}, \mathbf{1}/n), \quad (9)$$

where $\mathcal{L}_{IQN}$ denotes the IQN loss function applied to the noisy transition datapoint $T_i = (s_i, a_i, r_i, s_{i+1})$ from replay buffer of size n . The IQN model uses parameters $\theta \in \mathbb{R}^d$, while $\mathbf{P} = (p_1, \dots, p_n)$ denotes noise-characterizing density ratios satisfying $\sum p_i = 1$ and $p_i > 0$. $d(\mathbf{P}, \mathbf{1}/n)$ quantifies the difference between $\mathbf{P}$ and the uniform probabilities $\mathbf{1}/n$.

To adapt the DRO to big data and large parameter scenario, we focus on the uncertainty set $\mathcal{Q}$, which is partitioned into J subgroups that represent the subgroups of transition data with different noise types. Based on this replay buffer partition, we propose an efficient gradient-based learning approach to solve the optimization problem.

Given identical noise types per subgroup as detailed in Section 4.1, we implement independent sampling across all replay-buffer subgroups. Following Group DRO [Liu *et al.*,

2025], we optimize for worst-group performance by considering an ambiguity set formed by a finite set of subgroup distributions and minimizing the maximum subgroup loss. While this worst-group objective improves robustness, it can be overly conservative and may under-utilize informative samples from non-worst groups. To further improve both comprehensive robustness and data utilization efficiency beyond the worst-case subgroup, we compute the optimization gradient as a convex combination of gradients from all noisy subgroups. The optimization gradient in DRIQN is used to replace the original gradient during the backpropagation through the last quantiles layer of the IQN network.

We formalize the inner term of DRO problem Eq. (9) with a partitioned uncertainty set $\mathcal{Q}$. For simplicity, we construct $\mathcal{Q}$ through considering limited J kinds of noise, i.e. $\mathcal{Q} := \{q_1, q_2, \dots, q_J\}$.

$$\sup_{\mathbf{P} \in \mathbb{R}^J, q \in \mathcal{Q}} \sum_{j=1}^n p_j \mathbb{E}_{(T_j) \sim q_j} [\mathcal{L}_{IQN}(T_j)] - d(\mathbf{P}, \mathbf{1}/J). \quad (10)$$

To ensure a fair comparison of natural-condition-induced noise types, we assume equiprobable occurrence for all categories, which yields $d(\mathbf{P}, \mathbf{1}/J) = 0$ in the final term. So the inner term of DRO will be expressed as:

$$\sup_{q \in \mathcal{Q}} \mathbb{E}_{T \sim q} [\mathcal{L}_{IQN_q}(T)]. \quad (11)$$

Now we go back to the complete DRO minimax term Eq. (9). To solve the problem, we aim to minimize the expected loss over the uncertainty set $\mathcal{Q}$ following:

$$\min_{\theta} \max_{1 \leq j \leq J} \frac{1}{c_j} \sum_{k=1}^{c_j} \mathcal{L}_{IQN}(T_{jk}), \quad (12)$$

where J is the number of noise subgroups, c_j is the number of samples in subgroup j that has unique noise type, T_{jk} refers to the k -th transition data in the j -th subgroup. Our method provides a more scalable and efficient way to obtain the gradient descent direction by solving optimization problem Eq. (12) through learning the supremum of the set of loss functions with respect to each noise subgroup.

As the optimization problem described in Eq. (12), the right side noted as $f_j(\theta) = \frac{1}{c_j} \sum_{k=1}^{c_j} \mathcal{L}_{IQN}(T_{jk})$ during the iteration m can be linearized to achieve convex approximation, therefore we rewrite the right side as $f_j(\theta_m) + \langle \nabla f_j(\theta_m), \theta - \theta_m \rangle$.

Since the inner term for max operation has been smoothed by linearization, to ensure finding the stable descent direction during the min stage when $\max_{1 \leq j \leq J} f_j(\theta_m) + \langle \nabla f_j(\theta_m), \theta - \theta_m \rangle$ is not strictly convex with respect to parameter θ . Thus, we add a regularization term $\|\theta - \theta_m\|_2$. The gradient descent direction is set as $\delta := \theta - \theta_m$, we obtain the equivalent form of the DRO problem:

$$\min_{\delta, \kappa} \|\delta\|_2 + \kappa, \quad (13a)$$

$$\text{s.t. } f_j(\theta_m) + \langle \nabla f_j(\theta_m), \delta \rangle \leq \kappa, \quad \forall 1 \leq j \leq J. \quad (13b)$$

Theorem 1 (Equivalent dual quadratic program). *Let $\theta_m \in \mathbb{R}^d$ be the model parameter at iteration m and define the subgroup losses $f_j(\theta)$, $j = 1, \dots, J$. Denote $G_m := \nabla_{\theta} f(\theta_m) \in$*

$\mathbb{R}^{J \times d}$, whose j -th row is $\nabla_{\theta} f_j(\theta_m)^\top$, and consider the quadratic program in Eq. (13). The unique optimal descent direction of this problem is a convex combination form:

$$\delta_m^* = -G_m^\top \lambda^*, \quad \lambda^* \in \Delta_J := \left\{ \lambda \in \mathbb{R}_{\geq 0}^J \mid \sum_{j=1}^J \lambda_j = 1 \right\}, \quad (14)$$

where λ^* solves the dual quadratic program

$$\min_{\lambda \in \Delta_J} \frac{1}{2} \lambda^\top G_m G_m^\top \lambda - \lambda^\top f(\theta_m). \quad (15)$$

Hence δ_m^* is a linear combination of all subgroup gradients at iteration m .

The formulation preserves the Bellman operator’s monotonicity and γ -shift invariance, ensuring that value evaluation is well-posed for any stationary policy. Building on this foundation, Theorem 1 (proof adapted from [Shen *et al.*, 2022]) is detailed in Section A of Supplementary Material) establishes that the original constrained minimax problem can be solved by an equivalent dual quadratic program of dimension J (the number of noise subgroups). This equivalence allows the optimization to be reduced from searching over high-dimensional vectors to finding a linear combination of subgroup gradients, incurring a computational complexity of $O(J^3)$. Crucially, the resulting convex-combination coefficients serve as a weight vector that adaptively allocates importance across subgroups. This implies that DRIQN leverages a weighted combination of subgroup gradients rather than discarding all but the worst-case sample. With bounded mini-batch gradients of $\max_{1 \leq j \leq J} f_j(\theta_m) + \langle \nabla f_j(\theta_m), \theta - \theta_m \rangle$, the stochastic convergence rate is $O(T^{-1/2})$ [Hazan *et al.*, 2006], where T is the number of iterations.

Remark. As characterized in Theorem 1, standard IQN updates suffer from gradient conflict under heterogeneous noise, where gradients from benign and hazardous subgroups oppose each other (i.e., $\langle \nabla f_j, \nabla f_k \rangle < 0$). This cancellation drives the learned quantiles toward the average return distribution, causing the agent to significantly underestimate the tail risks associated with worst-case scenarios.

4 Experiments

4.1 Experimental Settings

Simulation Environment. Following recent work, we evaluate robustness in simulation, as real-world marine experiments are prohibitively costly. To better approximate real-world navigation under uncertainty, we evaluate our method in two perturbation-prone simulation settings. We first adopt a risky marine environment for USV navigation based on prior work [Lin *et al.*, 2023]. To further assess generalization across robot types and navigation contexts, we additionally consider a UAV task in an indoor cluttered environment [Liu *et al.*, 2022]. In both settings, we use a purely sensor-based navigation scheme without any prior knowledge. For the marine setting, we randomize eight vortices and six obstacles to create challenging layouts under severe noise. We train the models for 1.5M time steps over nine seeds and evaluate every 10,000 steps on 15 pre-randomized environments, resetting the environmental layout each episode. For the UAV

setting, we keep six obstacles and follow the original experimental protocol with 200,000 training frames.

Noise and Grouping Settings. The critical challenge lies in maintaining the robustness of DistRL under noisy observation. In order to narrow the gap between simulation and real sensing, we ground our noise modeling in prior physical studies [Koenig and Howard, 2004; Laconte *et al.*, 2021] and superimpose ubiquitous non-synthetic sensor noise processes, specifically Gaussian and Poisson noise. To assess DRIQN’s multi-noise robustness, we incorporate extra salt-and-pepper noise (simulating extreme value fluctuations) and occlusion noise (emulating partial sensor failures) in Section 4.4. The noise intensity is treated as a tunable hyperparameter and is empirically set to $\{0.6, 0.4, 0.2\}$ to systematically assess performance degradation under different perturbation levels; for the cluttered setting, we fix the intensity at 0.2. The sensor value range and noise intensity jointly determine the resulting distributions. As our focus is enhancing robustness in online DistRL training via DRO, we adopt a simple, practice-aligned grouping scheme based on collection periods with known noise types rather than clustering: data collected within the same period forms one subgroup, reflecting that noise conditions naturally shift across time and regions. This DRO framework is agnostic to the specific noise law, enabling extensions to other settings, including safety analysis under adversarial perturbations.

Baselines. We compare DRIQN with the following representative methods in autonomous navigation: (1) Classical observation-based planning methods: Artificial Potential Field (APF) method [Fan *et al.*, 2020] and Bug Algorithm (BA) method [Lumelsky and Skewis, 1988]; (2) Standard Q-learning method in DRL: DQN [Mnih *et al.*, 2015]; (3) Risk-sensitive method in DistRL: IQN [Dabney *et al.*, 2018].

Hyperparameter Settings. For neural networks with three or more layers, we implement gradient substitution exclusively in the final layer instead of the whole layers, inspired by established computer vision research on gradient manipulation [Shen *et al.*, 2022]. The comparative analysis is detailed in Section 4.5. Additionally, to adapt the gradients computed by the DRO to the training of all layers except the final one, we adapt the learning rate decay of these layers starting from 0.0001 to 0.000001 and appropriately shrink the calculated gradients. The implementation builds upon previous work mentioned in Section 4.1, with other hyperparameters configured in accordance with their respective network settings. The experiments were completed in parallel using six A6000 GPUs and AMD EPYC 7542 CPU.

4.2 Metrics and Strategy

Metrics. To evaluate the performance of our approach, we employed six metrics encompassing performance, safety, and efficiency: success rate, collision rate, timeout rate, final reward, average time consumed, and average energy consumed. Average time and energy are calculated based solely on data from successful episodes. Given the physical interpretation of these metrics, path smoothness can be indirectly quantified via the resulting time and energy costs.

Strategy. We compare a standard greedy strategy that selects the action with the highest expected return with the adaptive

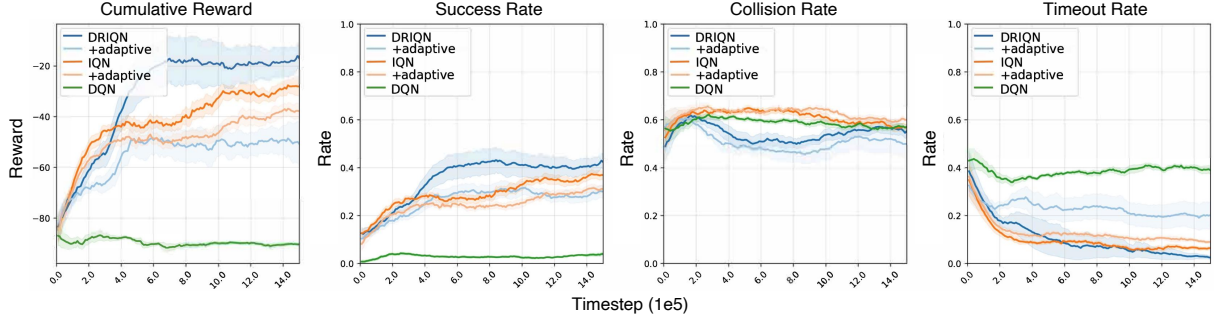


Figure 3: Online evaluations of the marine environment compare three learning-based models and two distributional RL strategies. Performance, safety, and convergence are assessed via three key metrics with cumulative reward curves. Implementation details appear in Section 4.1. Due to space limit, the other two figures beyond noise level 0.6 are in Supplementary Material, Section B.

Method	Strategy	SR $\uparrow$			CR $\downarrow$			TR $\downarrow$			FCR $\uparrow$			AT $\downarrow$			AE $\downarrow$		
		0.6	0.4	0.2	0.6	0.4	0.2	0.6	0.4	0.2	0.6	0.4	0.2	0.6	0.4	0.2	0.6	0.4	0.2
APF	–	0.11	0.25	0.03	0.81	0.55	0.53	0.08	<u>0.20</u>	0.44	–	–	–	206.68	349.32	547.89	320.96	654.48	1071.70
BA	–	0.22	0.11	0.26	0.71	0.46	0.33	0.07	0.43	0.41	–	–	–	185.76	482.98	462.97	251.11	691.93	691.57
DQN	Greedy	0.04	0.03	0.02	<u>0.57</u>	0.57	0.57	0.39	0.40	0.41	-90.30	-90.11	-89.54	541.27	550.35	560.52	858.39	779.78	832.67
IQN	Greedy	<u>0.37</u>	<u>0.50</u>	<u>0.61</u>	<u>0.57</u>	<u>0.45</u>	<u>0.37</u>	<u>0.06</u>	0.04	<u>0.02</u>	-28.16	-14.87	13.47	144.89	123.75	61.25	196.73	181.63	92.18
	Adaptive	0.31	0.36	0.46	0.60	0.51	0.38	0.09	0.13	0.16	-37.93	-38.97	-48.66	185.75	230.00	274.27	247.34	296.00	358.99
DRIQN	Greedy	0.42	0.55	0.63	0.55	0.42	<u>0.37</u>	0.02	0.04	0.01	-17.11	3.14	26.87	93.51	90.11	31.82	141.67	114.15	53.61
	Adaptive	0.30	0.39	0.33	0.50	0.46	0.49	0.20	0.15	0.18	-50.84	-36.16	-45.90	283.85	226.07	272.25	392.90	257.17	401.26
Improvement (%)		13.51	10.00	3.28	12.28	6.67	-12.12	66.67	–	50.00	39.24	121.12	99.48	35.46	27.18	48.05	27.99	37.15	41.84

Table 1: **Overall performance comparison** under three different noise levels (0.6, 0.4, and 0.2) related to the variances of the noise distributions. The best results are highlighted in **bold** and the second best results are highlighted with an underline. “Improvement” denotes the percentage increase (positive metrics) or decrease (negative metrics) over the second best models. For brevity, we use corresponding abbreviations for the metrics in this and all subsequent tables: success rate (SR), collision rate (CR), timeout rate (TR), final cumulative reward (FCR), average time consumed (AT), and average energy consumed (AE).

IQN strategy of [Lin *et al.*, 2023], which adjusts the CVaR threshold based on real-time risk estimates (proxied by distance to the nearest obstacle).

4.3 Performance Comparison

Online evaluation results in Figure 3 (reported for the marine setting, which adopts the more challenging training protocol) are smoothed using an Exponential Moving Average to highlight learning dynamics; shaded regions denote the standard deviation over nine seeds under high noise intensity. Through multiple trials, the shaded regions demonstrate the significance of DRIQN’s performance. DRIQN adds an overhead of only 0.018ms per update and outperforms the runner-up.

For USV navigation, the qualitative trajectory results of RL agents are visualized in Figure 4. Key performance illustrated in Table 1 are further summarized as follows. The improvements are statistically significant (e.g., task-critical metrics Success: $p = 0.0093$; Collision: $p = 0.0380$).

- *DRIQN vs. Baseline Methods*: As a DistRL approach enhanced by DRO-based gradient substitution, DRIQN achieves superior success / collision rates and impressive efficiency compared with other traditional physical

methods and RL algorithms across noise intensities, particularly under high noise variance.

- *DistRL vs. Standard RL*: DistRL methods consistently outperform standard deep Q-learning. DQN exhibits high timeout rates and resource consumption due to passive collision avoidance, stemming from its reliance on expected returns that are less informative in noisy settings. In contrast, DistRL explicitly models return distributions, yielding stronger noise robustness.
- *Greedy vs. Adaptive*: Both IQN and DRIQN achieve higher success rates with greedy action selection. Adaptive strategies degrade under noise, as perturbed value distributions undermine their decision rule.
- *DRIQN Efficiency*: Under stringent noise-corrupted conditions, greedy DRIQN maintains the lowest timeout rate while reducing energy and time consumption, optimally balancing safety and efficiency.

The additional drone navigation experiments in Table 2 further validate DRIQN’s superiority under observational noise, supporting its generalization across robot classes and navigation environments. Qualitative trajectories from IQN and DRIQN are shown in Figure 5.

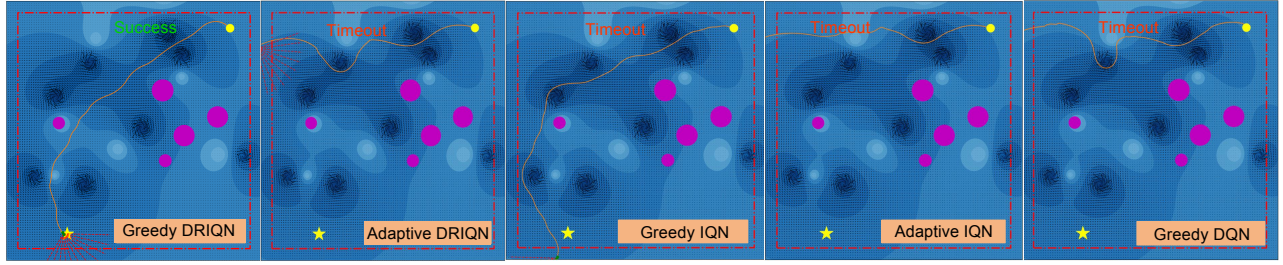


Figure 4: **Qualitative trajectory results** (noise variance level = 0.6). **Yellow circle**: start position, **yellow star**: the goal, **Magenta circles**: static obstacles. The greedy approach avoids risk-sensitive adjustments triggered by noisy observations, unlike the adaptive strategy. Aligning with Table 1, the adaptive strategy exhibits higher timeout rates and frequent deviations from complex conditions to mitigate risk.

Method (Greedy)	SR $\uparrow$			FCR $\uparrow$		
	0.6	0.4	0.2	0.6	0.4	0.2
IQN	0.20	0.35	0.27	-16.63	-3.20	-7.85
DRIQN	0.43	0.41	0.46	0.85	0.50	5.06

Table 2: Drone navigation performance under all noise levels, evaluated with greedy action selection on 100 randomly generated noisy environments with six obstacles each.

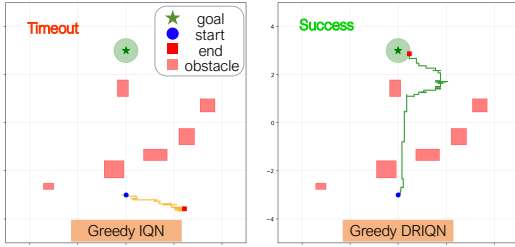


Figure 5: Qualitative drone trajectories generated by IQN (left) and DRIQN (right) under strong noise level 0.6.

4.4 The Analysis of “Noise Types”

To rigorously evaluate DRIQN’s safety in multi-noise conditions, we augment natural noise conditions with salt-and-pepper noise (simulating sensor malfunction) and occlusion noise (emulating data loss scenarios) for faulty sensors, setting a fractionally lower noise intensity across all sufficient perturbations for intensity-diversity balance. The strategy is rigged to greedy rather than adaptive for more competitive performance in this task, as empirically demonstrated earlier. Evidenced in Table 3, DRIQN’s optimization across heterogeneous noise subgroups demonstrates superior handling of complex perceptual patterns and observational robustness over baselines. We also try to replace the original grouping with success and trajectory length grouping. New partitioning drops the Success by 73% and increases the Collision by 4% at 0.6 intensity, justifying rationality of the original noise type based grouping.

4.5 The Analysis of “Gradient Substitution”

To examine our last-layer gradient substitution setting for DRIQN, we introduce the variant DRIQN-W featuring whole-network (4-layer) gradient replacement for compari-

Model	SR $\uparrow$	CR $\downarrow$	TR $\downarrow$	FCR $\uparrow$	AT $\downarrow$	AE $\downarrow$
APF	0.12	0.80	0.08	–	211.83	351.38
BA	0.14	0.82	0.04	–	134.65	198.20
DQN	0.11	0.50	0.39	-80.02	507.04	798.70
IQN	0.54	0.44	0.02	-3.32	75.27	108.54
DRIQN	0.58	0.41	0.01	12.45	45.10	73.30

Table 3: Robustness comparison under the multi-noise setting, where natural observation noise is combined with salt-and-pepper noise and occlusion noise.

Model	Strategy	SR $\uparrow$	CR $\downarrow$	AT $\downarrow$
DRIQN	Greedy	0.42	0.55	93.51
	Adaptive	0.30	0.50	283.85
DRIQN-W	Greedy	<u>0.22</u>	<u>0.54</u>	<u>357.87</u>
	Adaptive	0.20	0.55	379.50

Table 4: Architectural comparison of “gradient substitution” under high noise variance: Whole-network substitution (DRIQN-W) versus final-layer substitution (DRIQN).

son. As evidenced in Table 4, last-layer substitution significantly outperforms whole-network gradient replacement in both success rate and computational efficiency while ensuring better safety. This advantage stems from retaining deep networks’ representational capacity while improving robustness via streamlined optimization, even under extreme noise. Learning curves (Figure S2 in Supplementary Material) show that last-layer substitution converges faster and more stably than full-network modification.

5 Conclusion

In this paper, we propose DRIQN, a robust and efficient DistRL policy for path planning under observation noise. DRIQN combines flexible gradient substitution with DRO-guided gradient computation to improve implicit-quantile DistRL under heterogeneous noise. Experiments show consistent gains over strong baselines and validate robustness across tasks and vehicle types. We further find that greedy action selection is more reliable than distance-based adaptation under noisy observations, highlighting the fragility of hand-crafted physical cues. Future work will extend DRIQN to continuous control and real-world deployment.

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